The Particle Theory of Matter

1. All matter is made of tiny particles. These particles are always in motion (moving).

As the matter is heated the particles move faster and further apart.

State of matter	Movement of particles	Forces holding particles together	How close particles are together	Diagram
Solid	Particles v_____ in fixed positions	Very strong	Close	
Liquid	Particles able to roll over each other	Strong	Close	
Gas	Particles move _____ _____	Weak	Far apart	

2. Questions

a. In which state of matter do Particles move fastest?

b. How can you make particles move faster?

c. What happens to the density of substances when they are cooled?

Types of Change in Matter

3. There are two types of change seen with matter:

a. Physical change. Here there is a change in state or shape, but new substance is made. Physical changes are normally reversible.

b. Chemical changes. Here a new substance is made. It is normally impossible to reverse a chemical change.

Change	Is it physical or chemical?	Was a new substance made?
Ice melting		
Paper burning		
Wax melting		
Food rotting		
Water boiling		
Food digesting in the gut		
A ball of wet clay is squashed into a flat sheet		
Iron metal reacts with oxygen gas to form iron oxide(rust)		

4. Sodium chloride (NaCl) solution is mixed with a silver nitrate ($AgNO_3$) solution, white silver chloride (AgCl) is produced, leaving behind sodium nitrate ($NaNO_3$).

   a. Write a word equation for this reaction:

b. Write a formulae equation for this reaction:

c. What is corrosion?

d. What do we call corrosion to iron?

5. The reaction can be seen as iron (Fe) reacting with oxygen (O_2) to produce iron (III) oixide (Fe_2O_3).

   a. Write a word equation for this reaction:

   b. Write a formulae equation for this reaction:

6. Draw a molecule with 3 atoms:

7. Draw a compound with 4 atoms: